Lingala Vedamsh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.10
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	97.70%
Narayana Olympiad School Tarnaka	Class X (Board of Secondary Education Telangana)	2021	10.00%

WORK EXPERIENCE

Data Science Intern - Sprih

ESG Reporting Agent

- Built a **multi-agent** ESG sustainability reporting system using **LangChain Deep Agents**, automating company **web research** and **report drafting** that saves hours of manual work.
- Engineered an **MCP server** exposing the agent as a callable tool to external AI platforms (e.g., **Claude Cowork**), using an async task-based architecture to handle **long-running research jobs** within platform timeout constraints.
- Built an **MS Word plugin** (Vue, Office.js) enabling in-document AI interaction for easier report editing; crafted **OOXML**-based targeted-edit generation to preserve native formatting for inline, reviewable AI revisions.
- **Optimized research coverage** from **~30% to ~65%** on a priority-tiered evaluation dataset via **LangSmith** through **hypothesis-driven** prompt iteration.

Database Agent

- Built a **natural language-to-SQL** agent using **Claude Agent SDK**, answering queries across an ESG database spanning **15K+ companies**, and generating Excel reports directly from retrieved data
- Designed agent harness with **skills**, schema-grounded context via **table and column-level comments**, and **security guardrails** and evaluated **trade-offs** against other agent architectures such as **Databricks**.

PROJECTS

Medical Review Sentiment Analysis

- Built an NLP pipeline for multi-class sentiment classification (negative/neutral/positive) on Drug.com medical reviews, using a **transformer-based RoBERTa** model.
- Applied **LoRA (Low-Rank Adaptation)** to **fine-tune** only a small fraction of the model's parameters, slashing compute and memory needs while preserving full fine-tuning performance.
- Performed text preprocessing, exploratory data analysis, and feature engineering, and also **benchmarked** results with traditional ML and DL models.

Disease Prediction using ECG Signal Analysis

- Built a **CNN-LSTM** classification model to predict cardiac disease types from ECG signals.
- Designed an **unsupervised CNN Autoencoder** for **anomaly detection**, flagging abnormal heartbeats via reconstruction errors to support early diagnosis.

BitSqueezer

- Built a **C++ file compressor** using **Huffman coding** for lossless compression and decompression of binary files.
- Designed a deterministic **Huffman tree** with a **min-priority queue** and **byte-value tie-breaking**, ensuring reproducible tree construction across compression and decompression.
- Implemented bit-level code packing, binary-safe frequency counting, and a self-describing file header for standalone decompression.

SKILLS

Languages: C/C++, Python, SQL, MATLAB

Libraries: STL, Pandas, NumPy, Seaborn, Matplotlib, Scikit-Learn, TensorFlow

Frameworks: LangChain, LangGraph, Deep Agents, Claude Agent SDK

Tools: GitHub, Google Colab, Jupyter, VS Code, LangSmith, Docker

Concepts: Agentic AI, MCP, RAG, Machine Learning, Deep Learning, Data Structures and Algorithms, SQL

RELEVANT COURSES

CSE/AI Courses:

Data Structures & Applications, Algorithms and Data Structure Lab, Introduction to Modern AI, Artificial Intelligence

Math Courses:

Probability and Random Variables, Probability and Random Processes, Statistics, Linear Algebra, Calculus-I, Calculus-II

POSITIONS OF RESPONSIBILITY

MedTech Domain Head - Elektronica Club (25-26)

- Led the **Indian Air Force** Frostbite Prevention Socks Project, coordinating with IAF personnel and professors.
- Led MedTech initiatives, including an EMG-assisted prosthetic arm prototype; conducted sessions for 50+ students.

Core - Elektronica Club (24-25)

- Built Arduino-based projects and developed biosignal analysis models for ECG and EEG data.

EXTRACURRICULAR

- National Cadet Corps C Certificate holder.
- Cricket enthusiast, from hostel rooms to the field.